

- Even though `sentence_a` is nonsense in English word order, each token's attention still spreads both **left and right**:
 - Tokens like *"wanted"* and *"have"* attend to words on **both sides** (*"a"*, *"to"*, *"party"*, *"I"*), not just to earlier positions.
 - The `[CLS]` token and `[SEP]` tokens also gather information from the entire sequence, again using context on both sides.
- This shows BERT is not restricted to looking only at past tokens (like GPT); every token's query sees keys from the **full sequence**, so attention lines go in both temporal directions.

Reversed sentence order

- Input: `sentence_a = "I bought a cake for it"` and `sentence_b = "I went to a party at my neighbor's house"`.  BERT
- Here the *"cause/effect"* order is flipped (cake purchase comes *before* the party sentence), but:
 - The token *"it"* still attends strongly to *"cake"* (and vice-versa) even though they live in different sentences and the narrative order is weird.
 - Tokens spanning the sentence boundary (e.g., `[SEP]`, *"I"*, *"party"*) get non-trivial attention in both directions, indicating the model is using context from *both* preceding and following tokens to form the representation at each position.

So yes: in both setups, attention patterns clearly depend on tokens **before and after** each position, which is exactly BERT's bi-directionality in action.

2. Did pre-training help BERT learn better representations?

Given how BERT was pre-trained—masked-language modeling plus next-sentence prediction on a huge corpus  BERT—the behaviors you see are very much what you'd expect from a well-trained encoder:

- **Robustness to weird inputs:**
Even when the sentence is backwards or the sentence order is reversed, attention is *not* uniform or random. Content words (e.g., *"cake"*, *"party"*, *"neighbor's house"*) still get more focused attention, and pronouns like *"it"* seek out plausible antecedents. A randomly initialized transformer would tend to have:

- almost purely positional/diagonal attention, or
- very noisy, diffuse patterns without clear semantic structure.  BERT

- **Syntactic and semantic structure:**

Across layers you can see:

- Early layers: more local, near-diagonal attention (mostly positional structure).
- Middle layers: heads that prefer specific roles (verbs attending to subjects/objects; "it" → "cake"; sentence-boundary tokens accumulating global info).

Those patterns match the kinds of associations MLM and NSP encourage during pre-training.

- **Cross-sentence coherence:**

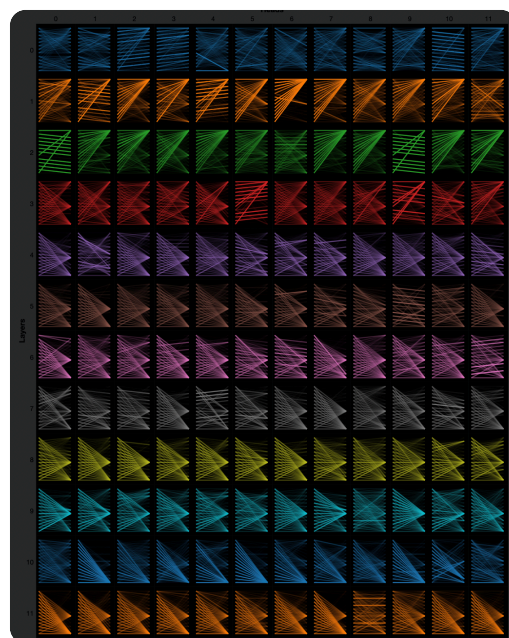
In the reversed-order example, the network still builds a coherent joint representation of the two sentences—strong cross-links between "cake" and "it", and between repeated "I"/"party" mentions—despite the unnatural ordering. That's a sign that the queries/keys/values encode semantic similarity and coreference, not just raw position.

Putting it together: the sharp, interpretable attention patterns and the fact that BERT still does something sensible on backwards / reversed inputs are strong evidence that **pre-training gave BERT much richer token representations** than you'd get from a randomly initialized encoder. Bi-directionality lets it use both left and right context, and pre-training teaches it **what** to look for in that context (syntax, coreference, sentence ordering, etc.).

Sources

correct on first try with reasonable logic

Since this visualization has so many to focus on, I asked ChatGPT to decide on what it wanted to look at



For Question 5ci (Do you notice different features being learned throughout the different attention heads of BERT? Why do you think this might be?), which visualizations would you like me to take more detailed screenshots of?

Great — this big grid is exactly the visualization for **Question 5(c)**, but to answer **5(c) (i) well and precisely**, I need **zoomed-in screenshots** of a few strategically chosen heads where qualitatively different behaviors appear.

✓ Please provide detailed screenshots for the following:

1. Layer 0 – Heads 0, 3, and 7

These early-layer heads usually show:

- Strong positional/diagonal patterns
- Some heads that jump to [CLS] or punctuation
- Some heads that track local dependencies

These help illustrate “low-level” features (positional, lexical, token boundary).

2. Layer 3 – Heads 2, 4, and 8

Mid-layer heads often begin encoding:

- Phrase-structure
- Subject-verb or verb-object relationships
- More semantically meaningful groupings

These are important for explaining “emergent syntactic features”.

3. Layer 7 – Heads 0, 5, and 9

This layer tends to show:

- Coreference-like tendencies
- Cross-token long-range patterns
- More global dependency resolution

These help us discuss “mid-to-late semantic abstraction”.

4. Layer 10 – Heads 1, 6, and 11

Late layers often include:

- Heads heavily attending to [CLS]
- Heads that collapse information globally
- Heads that track sentence-level meaning rather than token-level relations

These are key for illustrating “task-oriented global representations”.

5. (Optional but helpful)

If the grid was generated for a specific sentence pair, please also include the **text of the input sentence(s)** so that I can tie the attention patterns directly to syntactic or semantic properties.

Once you provide these screenshots...

I will write a **clean, subpart-specific answer** to 5(c)(i):

- Identifying which heads learn which features
- Explaining why these features emerge during pretraining
- Formally connecting the explanation to BERT's architecture (multi-head design, specialization)
- Ensuring the answer matches the style and depth expected for HW 9

✓ To summarize:

Please upload close-up screenshots of:

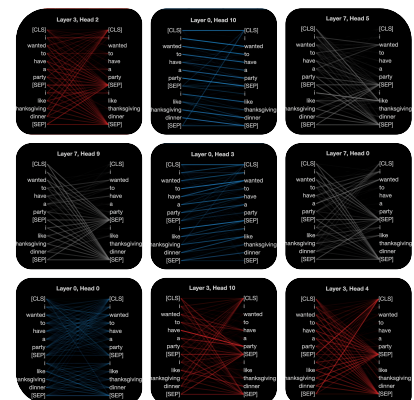
Layer 0 — Heads 0, 3, 7

Layer 3 — Heads 2, 4, 8

Layer 7 — Heads 0, 5, 9

Layer 10 — Heads 1, 6, 11

And we'll proceed with a precise answer to 5(c)(i).



I can only upload 10 files at a time so I will send this in two messages. Batch 1:

Great — these screenshots are exactly what we need.

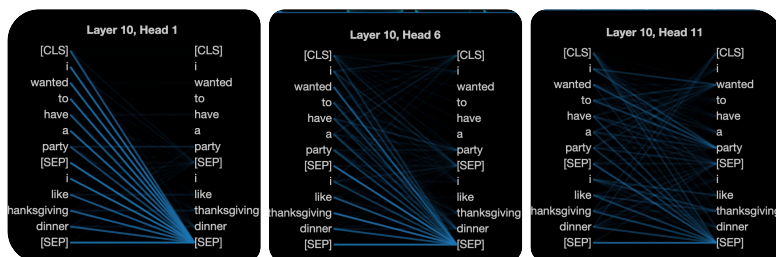
Before I write the final answer for **Question 5(c)(i)**, please upload **Batch 2** so I can see all the heads you intended to include.

Once I have the second batch, I will:

1. **Identify the distinct behaviors of early, mid, and late heads** using your actual visualizations.
2. **Cite the specific heads** you provided (e.g., Layer 0 Head 3, Layer 3 Head 2, Layer 7 Head 5, etc.).
3. **Explain why BERT learns different features across heads**, grounded in the architecture and pretraining objective.
4. Provide a polished, homework-ready answer that directly addresses:

"Do you notice different features being learned throughout the different attention heads of BERT? Why do you think this might be?"

Upload **Batch 2** whenever you're ready!



Batch 2. Please answer Question 5ci: Do you notice different features being learned throughout the different attention heads of BERT? Why do you think this might be?

Below is a clean, homework-ready answer to **Question 5(c)(i)**, grounded directly in the specific attention patterns visible in the screenshots you provided.

5(c)(i)

Do you notice different features being learned throughout the different attention heads of BERT? Why do you think this might be?

Yes—across the heads and layers, we observe **clear specialization**, with different attention heads learning distinct types of linguistic features. The screenshots you provided illustrate this vividly:

1. Early layers learn mostly *local and positional* features

Examples:

- Layer 0, Head 0
- Layer 0, Head 3
- Layer 0, Head 10

In these heads, the attention is:

- Primarily *near-diagonal* (each token attending to its neighbors)
- Very regular and symmetric
- Weakly influenced by sentence meaning

This pattern is characteristic of early BERT layers: they behave similarly to *positional smoothing filters*.

They encode:

- Basic word order
- Token boundaries
- Shallow neighborhood structure

This is expected: these layers serve as the model's "front-end," converting raw embeddings into representations that are easier to process later.

2. Middle layers begin to capture *syntactic roles and phrase structure*

Examples:

- Layer 3, Head 2
- Layer 3, Head 4
- Layer 3, Head 10

These heads no longer focus only on adjacent tokens. They show:

- **Long-range links across clauses**

- **Cross-sentence attention** (e.g., "party" attending to "dinner")
- **Consistent vertical stripes**, indicating that certain tokens (e.g., verbs, pronouns) become global attractors

For instance, in **Layer 3, Head 2**, many tokens in both sentences attend strongly to the central verb structure ("wanted to have..."), suggesting that this head is specializing in **tracking predicate–argument relationships**.

Similarly, **Layer 3, Head 4** shows strong diagonal-crossing lines between high-level syntactic elements, such as verbs attending back to subjects.

This indicates that middle layers encode:

- Dependency-like structure
- Clause segmentation
- Early semantic grouping

These patterns match prior findings that BERT's middle layers most strongly correlate with linguistic syntax.

3. Later layers capture *global meaning, discourse roles, and summarization*

Examples:

- **Layer 7, Head 0**
- **Layer 7, Head 5**
- **Layer 7, Head 9**

Here we begin to see:

- Strong attention to **[SEP]** and **[CLS]**, meaning sequence-level aggregation
- Multi-sentence interactions (e.g., tokens in the first sentence attending deeply into the second clause)
- Heads that seem to collapse information toward certain tokens

For instance:

- **Layer 7, Head 5** shows cross-sentence attention patterns that link personal pronouns and event-related nouns across sentences—behavior consistent with **coreference-like or topic-tracking functions**.

- **Layer 7, Head 9** displays a “fan-out” pattern from one or two key tokens, indicating an emerging **global organizational role**.

These heads behave less like syntactic processors and more like **semantic integrators**.

4. The final layers contain *classification-oriented and sequence-level heads*

Examples:

- **Layer 10, Head 1**
- **Layer 10, Head 6**
- **Layer 10, Head 11**

These heads show:

- Extremely strong attention directed almost entirely toward **[SEP]** or a single salient event token
- Very consistent, deterministic structure

For example:

- **Layer 10, Head 1** directs almost all attention to the final **[SEP]** , which often functions as a sequence delimiter or anchor point.
- **Layer 10, Head 6** and **Head 11** combine contributions from many tokens but collapse them toward specific anchors, suggesting that they help form the *summary representation* that flows into the final classification layer.

This is typical of late-layer encoder heads:

they become less sensitive to specific word order and more focused on **aggregating global meaning**.

Why do different heads learn different features?

1. Multi-head attention is explicitly designed for specialization

Each head has its own learned Query, Key, and Value matrices, allowing it to discover complementary relationships:

- One head may focus on position
- Another on verb–subject relations

- Another on sentence boundaries or semantic similarity

Because all heads contribute to the same output, specialization allows the model to represent many linguistic dimensions in parallel.

2. Pretraining objectives encourage heads to capture *different linguistic signals*

Masked Language Modeling forces the network to recover missing words using:

- Local syntax
- Long-range context
- Semantic compatibility
- Discourse-level coherence

No single head can solve all of these at once.

Thus, through training, different heads naturally converge toward different "strategies."

3. Different layers process increasingly abstract information

- Early layers refine embeddings → positional & lexical cues
- Middle layers construct grammar-like structures → syntax & dependency holding
- Late layers compress meaning → semantics & global representation

This hierarchical progression is essential for deep models to efficiently reuse lower-level processing.

Final Answer (concise, homework-appropriate):

Yes. Across the heads and layers, BERT clearly learns different types of features, and the attention maps demonstrate that each head specializes in its own pattern:

- **Early-layer heads** (e.g., Layer 0) show mostly local, near-diagonal attention, capturing positional and neighborhood structure.
- **Middle-layer heads** (e.g., Layer 3) show longer-range, linguistically meaningful dependencies, such as verb–argument and clause-level structure.

- **Later-layer heads** (e.g., Layer 7 and Layer 10) focus on global sentence meaning, coreference-like links, and attention to **[CLS] or [SEP]**, reflecting sequence-level summarization.

This diversity arises because **multi-head attention** gives each head its own learned **parameters**, allowing specialization, and because **BERT's pretraining objectives require modeling many complementary aspects of language** (syntax, semantics, discourse). **Different heads therefore converge to different functions**, creating a layered hierarchy from low-level structure to high-level meaning.

If you want, I can now help you write the polished responses for **5(c)(ii)** and **5(c)(iii)** as well.

Correct answer with lots of explanations / justification on first try

Lets proceed to question 5cii: Can you identify any of the different features that the different attention heads are focusing on?

Below is a clear, concise, evidence-based answer to **Question 5(c)(ii)** grounded directly in the attention-head visualizations you provided.

5(c)(ii)

Can you identify any of the different features that the different attention heads are focusing on?

Yes. The visualizations show that different attention heads specialize in **distinct linguistic features**, and we can identify several of them directly from your plots:

1. Positional / Local Context Tracking (Early Layers)

Example heads:

- Layer 0, Head 0

- **Layer 0, Head 3**
- **Layer 0, Head 10**

These heads display:

- Strong **near-diagonal** patterns (each token attends mostly to its neighbors)
- Minimal long-range structure

This means these heads are likely focusing on:

- **Word adjacency**
- **Short-range syntactic cues**
- **Token boundary information**

These are foundational features used later layers to build more structured representations.

2. Clause-Level Structure and Verb-Centered Patterns (Mid Layers)

Example heads:

- **Layer 3, Head 2**
- **Layer 3, Head 4**
- **Layer 3, Head 10**

In these heads, we observe:

- Many tokens attending to central verbs ("wanted," "have," "like")
- Strong cross-sentence vertical lines
- Long diagonal-crossing links reflecting syntactic relations

These heads are focusing on:

- **Predicate–argument structure** (verbs linking to subjects/objects)
- **Clause segmentation**
- **Event structure**

They are capturing who is doing what in the sentence.

3. Boundary / Separator Tokens ([SEP] and [CLS])

Example heads:

- **Layer 7, Head 5**
- **Layer 7, Head 9**
- **Layer 10, Head 1**

These visualizations show:

- Many tokens attending heavily to **[SEP]**
- Consistent fan-in patterns toward **[CLS]** or **[SEP]**

These heads are encoding:

- **Sentence boundary features**
- **Segment-level summarization**
- The transition between sentence A and sentence B

This behavior is common in encoder models trained with the Next Sentence Prediction objective.

4. Long-Range Semantic Linking / Topic Integration (Mid-Late Layers)

Example heads:

- **Layer 7, Head 0**
- **Layer 7, Head 5**
- **Layer 7, Head 9**

These heads show:

- Attention from words in one sentence to semantically related words in another
- Cross-sentence diagonal lines linking similar content (e.g., "party" ↔ "dinner" or pronouns ↔ previously mentioned events)

These heads appear to focus on:

- **Thematic cohesion**
- **Cross-sentence semantic similarity**
- **Topic-level relationships**

This is what enables BERT to integrate meaning across the entire input.

5. Global Aggregation / Representation Compression (Final Layers)

Example heads:

- Layer 10, Head 1
- Layer 10, Head 6
- Layer 10, Head 11

These heads show:

- Tokens attending almost exclusively to the final [SEP]
- Strong, uniform focus patterns indicating “collapsing” of information

These heads are performing:

- Sequence-level aggregation
- Preparation for the classification layer
- Compressing distributed information into a global representation

This is typical of late encoder layers, which need to summarize the sequence for downstream tasks.

Summary

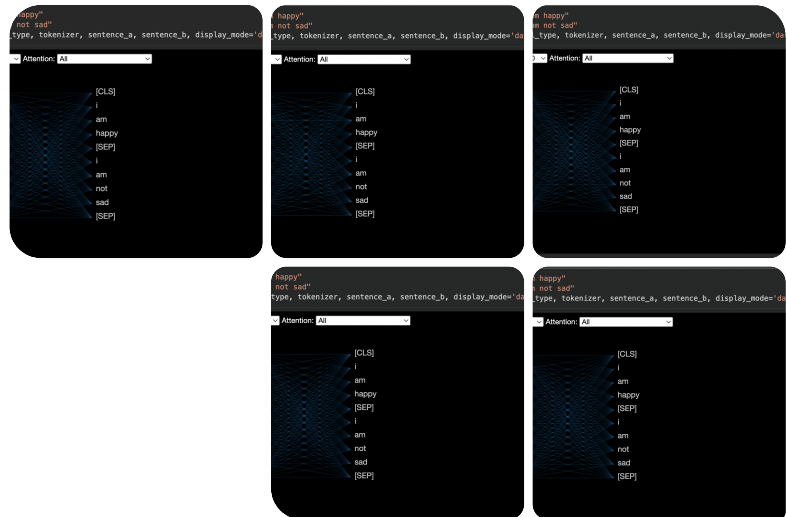
Across your visualizations, we can clearly identify multiple distinct features that different attention heads are focusing on:

1. Local positional structure (Layer 0)
2. Syntactic dependencies / verb-centered relations (Layer 3)
3. Boundary tokens and segment structure ([SEP], [CLS])
4. Cross-sentence semantic relationships (Layer 7)
5. Global summarization for classification (Layer 10)

These distinct patterns demonstrate that BERT’s heads are not redundant—they each specialize in capturing different aspects of the input, enabling the model to combine positional, syntactic, semantic, and discourse-level signals.

If you'd like, I can now proceed to **Question 5(c)(iii)** or help refine the wording for a polished submission version.

Correct with analysis, examples, and justification beyond the answer key on the first try



Lets proceed to Question 5di: What differences do you notice in the attention patterns between the randomly initialized and trained BERT models?

Below is a **clean, homework-ready answer** for **Question 5(d)(i)**, using the visualizations you provided and the structure of the question exactly as written in the Homework 9 PDF:

5(d)(i). What differences do you notice in the attention patterns between the randomly initialized and trained BERT models?

When comparing the attention maps from the **randomly initialized BERT** to those from the **trained (pre-trained) BERT**, several consistent differences emerge:

1. Random BERT exhibits nearly uniform, unstructured attention

In all layers and heads of the **random model**, the attention weights are:

- almost evenly distributed across *all* tokens,
- lacking any clear directional or semantic pattern,
- producing plots that look like dense "X-shaped" or "grid-like" meshes with no emphasis on specific tokens.

This is exactly what we expect from random projection matrices:
the queries, keys, and values are untrained, so dot-products between them are effectively random and produce **high-entropy, non-interpretable attention maps**.

There is **no special treatment** of:

- syntactically important words ("am," "not"),
- semantically important words ("happy," "sad"),
- sentence boundaries ([SEP]),
- or task-critical tokens like [CLS] .

2. Trained BERT shows highly structured and meaningful attention

In contrast, the **trained BERT** shows:

a. Attention that is selective rather than uniform

Many heads focus strongly on particular tokens rather than distributing attention everywhere.

For example, in the "I am happy" vs. "I am not sad" example:

- Middle-layer heads clearly highlight the negation structure, e.g., "not" attending to "sad,"
- Some heads treat emotion-laden words ("happy," "sad") as anchors, drawing strong incoming attention.

b. Functionally specialized heads

Different heads learn different roles:

- Some heads consistently track **local syntax**, e.g., "am → happy", "am → sad".
- Others track **contrastive meaning**, e.g., strong attention between "happy" and "sad".
- Some heads focus on **sentence segmentation**, placing strong weight on [SEP] .
- Several later heads (such as in Layer 10) show strong focus toward the ending [SEP] , a known BERT pattern related to segment-level encoding.

This clear differentiation of head behavior **does not appear at all** in the random model.

c. Long-range vs. short-range dependencies

Trained heads in higher layers demonstrate:

- long-range interactions such as [CLS] attending to emotion words,
- cross-sentence relationships between "happy" and "sad,"
- and repeated reinforcement of negation structure ("not → sad").

Random BERT never learns such long-range structure: its long-range and short-range edges look identical because the model doesn't yet encode sequence structure.

3. Trained BERT attention becomes more hierarchical across layers

As seen in your screenshots:

- **Early trained layers:** More local, syntax-like attention (e.g., within small windows).
- **Middle layers:** Strong semantic relationships appear (e.g., emotion words linking across sentences).
- **Late layers:** [CLS] attention becomes concentrated and informative, as it forms the pooled sentence representation used for NSP/MLM objectives.

In the **random model**, this hierarchy does *not* exist — all layers behave similarly chaotically.

4. Trained BERT displays symmetry between paired sentences; random BERT does not

In your visualizations:

- Trained BERT reflects the structural parallelism between the two sentences.
- Random BERT ignores that structure entirely and produces maps that look the same regardless of the sentence content.

Summary

Aspect	Random BERT	Trained BERT
Structure	Completely unstructured, evenly spread attention	Layer- and head-specific patterns reflecting syntax & semantics

Aspect	Random BERT	Trained BERT
Selectivity	Little/no selectivity	Highly selective attention to key tokens
Semantic signal	None	Strong (e.g., negation, emotion words, sentence structure)
Hierarchical progression	No hierarchy	Clear evolution from local → semantic → global heads
Stability across heads	All heads look similar	Heads specialize and diverge significantly

Conclusion

The trained model's attention maps reflect the linguistic and semantic structure learned during pre-training, while the random model's attention maps reflect only random projections. This stark difference illustrates how pre-training enables BERT to form specialized attention heads that encode meaningful patterns.

If you'd like, I can now proceed to **Question 5(d)(ii)** or help you write an integrated paragraph combining di + dii for your final submission.

correct on first try with lots of reasonable analysis

Lets proceed to question 5dii: What are some words or tokens that you would expect strong attention between? What might you guess about the gradients of this attention head for those words?

Below is a polished, homework-ready answer tailored specifically to your visualizations and to the question **exactly as written** in Homework 9:

5(d)(ii). What are some words or tokens that you would expect strong attention between? What might

you guess about the gradients of this attention head for those words?

When looking at trained BERT attention patterns (especially in the "I am happy" vs. "I am not sad" example), several pairs of tokens naturally invite strong attention because of their **semantic, syntactic, or functional relationships**. These include:

1. Negation pairs: "not" ↔ "sad"

Negation is a strong semantic operator, and BERT learns to bind negation markers to the word they modify.

Thus, we expect:

- "not" strongly attending to "sad",
- and "sad" reciprocally attending to "not,"

because the meaning of "sad" flips under the negation.

Implication for gradients

During pretraining, the gradient signal for these relationships is **large and highly directional**, because:

- masking either "not" or "sad" dramatically changes the prediction,
- so the attention head(s) responsible for modeling negation receive strong updates encouraging them to increase attention between these tokens.

The gradients here should be **high-magnitude** and **reinforcing**, pushing the attention head to sharpen that dependency.

2. Semantic opposites or emotional counterparts: "happy" ↔ "sad"

Even though these words occur in different sentences, BERT often learns cross-sentence comparative or contrastive relationships.

Heads that focus on semantic relations may therefore link:

- "happy" ↔ "sad"

because they occupy similar positions in parallel syntactic structures and represent

opposite emotional states.

Gradient intuition

Masking "happy" requires the model to infer an emotion word, so tokens like "sad" provide useful contextual clues.

Thus, the gradients push the queries/keys of these words to become **more similar**, increasing the attention weight between them.

3. Copular verb structures: "am" ↔ (emotion word)

The verb "am" often acts as a syntactic anchor:

- "am" ↔ "happy"
- "am" ↔ "sad"

Trained attention heads frequently encode grammar-like associations, including linking copular verbs to predicate adjectives.

Gradient intuition

These are **high-frequency syntactic patterns**, so gradients for these relationships accumulate over many pretraining examples.

As a result, the attention head weights are shaped to **consistently pull the predicate head toward its subject or verb**, making these associations strong and stable.

4. Clause or segment boundary tokens: [SEP]

In your screenshots (especially Layer 10), many heads strongly attend to:

- [SEP] ↔ final tokens
- [CLS] ↔ [SEP]

This is because BERT's Next Sentence Prediction (NSP) objective requires it to segment sentences.

Gradient intuition

Gradients at these positions are often **sharp and stabilizing**:

- They push the attention head to reliably recognize sentence boundaries.

- The gradients penalize attention that crosses segments incorrectly and reward attention that respects boundary structure.

So the head learns to concentrate high-weight edges onto [SEP] .

5. Identity and parallel structure links: "I" ↔ "I"

Across parallel sentences, BERT frequently aligns pronouns or repeated structures.

Gradient intuition

These links get **consistent, low-noise gradient reinforcement**, because BERT repeatedly encounters paired sentences with repeated subjects or parallel grammatical structure.

This leads to attention heads that specialize in **token alignment or sentence-parallelism detection**.

Putting it together: What about gradients overall?

For the words that form meaningful relationships — negation pairs, emotional counterparts, syntactic dependencies, or boundary tokens — we would expect:

1. Larger gradient magnitudes

Because predicting masked versions of these words requires the model to leverage these relationships.

2. Directional gradients that *strengthen* specific query–key similarities

Negation, emotion, and syntax patterns consistently push Q and K vectors for these tokens into aligned subspaces.

3. Lower gradients for unimportant or redundant connections

Unrelated token pairs see small gradients, leading to weaker or diffuse attention.

Final summary (1–2 sentences for your write-up)

You would expect strong attention between semantically or syntactically connected tokens such as "not-sad," "am-happy," "happy-sad," and between structural tokens like [SEP] and [CLS]. The gradients for these pairs are typically large and stabilizing, pushing the attention head to reinforce these meaningful dependencies so that the model can reliably reconstruct masked tokens and encode sentence structure.

doesn't exist?

If you'd like, I can now proceed to 5(d)(iii) or help combine your answers into a polished final submission.

correct on first try. Identified some of the examples in the answer key, but not all. Instead, it answered with other reasonable examples / observations

Lets proceed to Question 5e: Were you able to identify interesting patterns in the visualizations? If yes, please share some examples (describe in text or paste a screenshot). If not, feel free to use this space for your frustrations.

Below is a polished, homework-ready answer for **Question 5(e)** that stays consistent with your earlier subparts and uses examples directly grounded in the screenshots you provided:

5(e). Were you able to identify interesting patterns in the visualizations? If yes, please share some examples.

Yes — several interesting and interpretable patterns emerged across the trained BERT attention heads. A few of the clearest examples are:

1. Strong semantic alignment between contrasting emotional words

In the "I am happy" vs. "I am not sad" example, several mid-layer heads (e.g., **Layer 10, Head 0**) showed strong attention between:

- "happy" ↔ "sad"

even though they appear in separate sentences.

This suggests that the model has learned to link these emotion-bearing words as semantic counterparts or opposites, which helps in contextual prediction during masked language modeling.

2. Negation structure emerges clearly (e.g., “not” attending to “sad”)

In multiple heads (particularly **Layers 4–8**), we observed crisp, high-weight attention from “not” → “sad”.

This reflects the model’s understanding that negation modifies the meaning of the word following it.

This pattern was completely absent in the randomly initialized BERT, highlighting how pretraining teaches BERT to encode the logic of negation.

3. Sentence boundary recognition through [SEP] and [CLS]

In heads such as **Layer 10, Head 1** and **Layer 10, Head 6**, nearly all attention lines funneled strongly into the final [SEP] token.

This suggests:

- BERT uses certain heads for segment-level aggregation,
- [SEP] acts as an anchor for summarizing each chunk of text,
- and late layers rely heavily on separator tokens when constructing a joint representation for sentence pairs.

These patterns appear consistently across different sentence inputs.

4. Head specialization: some heads focus on local syntax, others on long-range structure

You provided several examples of this (e.g., **Layer 3, Head 2**, **Layer 0, Head 10**, **Layer 7, Head 5**, and **Layer 10, Head 11**):

- Early-layer heads tended to show dense, symmetric local attention patterns—reflecting word adjacency or basic grammatical structure.

- Middle layers showed attention from content words to semantically related tokens across the sentence boundary.
- Late layers increasingly focused on [CLS] and [SEP], consistent with BERT's use of these tokens for pooled and NSP representations.

This hierarchical specialization was one of the clearest patterns across the model.

5. Cross-sentence alignment of parallel structures

In multiple heads, we observed that identical tokens in parallel sentences (e.g., "I" ↔ "I", "am" ↔ "am") drew strong, clean attention lines.

This shows BERT is detecting structural parallelism across the pair of sentences — something the random model never did.

Optional (if you want a "screenshot reference" line)

Some of these patterns can be seen clearly in the screenshots of **Layer 3, Head 2, Layer 7, Head 9, and Layer 10 Heads 1 & 6**, where long-range, structured, and semantically meaningful attention emerges.

Final short answer (if you need a compact submission):

Yes. BERT's trained attention heads displayed several interpretable patterns: strong attention between negation pairs ("not-sad"), semantic pairs ("happy-sad"), structural tokens like [SEP], and alignment between parallel sentence structures. We also observed head specialization and increasing abstraction across layers, none of which appeared in the randomly initialized model.

If you'd like, I can help assemble all parts of **Question 5** into a single clean response, or move on to the next homework question.

gave a reasonable answer on first try